



2010 HSC Assessment Task 3 (Trial HSC)

- Working time – 3 hours.
- Commence each new question on a new sheet.
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- Board approved calculators may be used.
- All necessary working should be shown in every question.
- Attempt **all** questions.
- At the conclusion of the examination, bundle the sheets used in the correct order within this paper and hand to examination supervisors.

Class (please ✓)

- ☐ 12M21 – Mr Berry
- ☐ 12M22 – Mr Fletcher
- ☐ 12M31 – Mr Ireland
- ☐ 12M32 – Mr Lowe
- ☐ 12M33 – Mr Rezcallah

NAME: # PAGES USED:

Marker's use only.

[illegible]

Question 1 (12 Marks)	Commence a NEW page.	Marks
(a) Simplify $2\sqrt{11} \times 5\sqrt{11}$.		2
(b) Solve correct to 2 decimal places: $x^2 - 3x - 7 = 0$		3
(c) If $f(x) = \begin{cases} x - 2 & \text{for } x < 0 \\ x^2 & \text{for } x \geq 0 \end{cases}$ Evaluate $f(-1) + f(0) + f(2)$.		2
(d) Give the exact value of		
i. $\sin \frac{7\pi}{6}$.		1
ii. $\log_2 1$.		1
(e) A rectangle has sides of $\frac{1}{\sqrt{2}+1}$ cm and $\frac{1}{\sqrt{2}-1}$ cm. Find		
i. its area		1
ii. its perimeter in simplest surd form.		2

Question 2 (12 Marks)	Commence a NEW page.	Marks
(a) Differentiate with respect to x :		
i. $\frac{4x^2 - 3x + 2}{x}$.		2
ii. e^{3x^2-5} .		2
iii. $x^3 e^{-x}$.		2
iv. $\ln(\sin x)$.		2
(b) Explain, using calculus, why the curve $y = (3x - 2)^4$ has no points of inflexion.		2
(c) Solve for x , $ 3 - 2x \leq 5$.		2

- | | | |
|---|----------------------|--------------|
| Question 3 (12 Marks) | Commence a NEW page. | Marks |
| (a) What is the period and amplitude of the curve $y = 3 \cos 2x$? Sketch the curve for $0 \leq x \leq 2\pi$. | | 4 |
| (b) Write down a quadratic equation which is negative within the interval $-1 < x < 2$ and non negative for all other values of x . | | 2 |
| (c) Find the primitive of: | | |
| i. $\frac{2x}{x^2 + 1}$. | | 2 |
| ii. $\frac{2}{x} + 5e^x$. | | 2 |
| iii. $\frac{e^{2x}}{e^{2x} + 4}$. | | 2 |

- | | | |
|--|----------------------|--------------|
| Question 4 (12 Marks) | Commence a NEW page. | Marks |
| (a) The volume $V \text{ cm}^3$ of water remaining in a leaking bucket after t seconds is | | |
| $V = 2\,000 - 40t + 0.2t^2$ | | |
| i. How much water was there initially in the bucket? | | 1 |
| ii. How long does it take to empty the bucket? | | 1 |
| iii. How fast is the volume of water in the bucket decreasing when $t = 30$? | | 1 |
| (b) Find the value of $8 - 4 + 2 - 1 + 0.5 - 0.25 + \dots$. | | 2 |
| (c) Use Simpson's Rule with five function values to estimate the area bounded by the curve $y = \sqrt{x + 2}$, the x axis and the lines $x = 1$ and $x = -1$, correct to 2 decimal places. | | 3 |
| (d) Simplify $\frac{\sin^2 30^\circ}{1 - \sin^2 30^\circ}$. | | 2 |
| (e) Find the coordinates of the point on the curve $y = 2x^2 + 4x - 11$ where the tangent is parallel to the line $y = 8x + 9$. | | 2 |

Question 5	(12 Marks)	Commence a NEW page.	Marks
(a)	Find the equation of the normal to the curve $y = x \sin x$ where $x = \frac{\pi}{2}$.		3
(b)	For the quadratic equation $x^2 - (2 + k)x + 3k = 0$, find the value of k if		
	i. the sum of roots is 5.		1
	ii. the product of roots is 12.		1
(c)	Given that $\sin \theta = \frac{3}{4}$ and θ is acute, find the value of		
	i. $\tan \theta$.		1
	ii. $\sec \theta$.		1
	iii. $\operatorname{cosec}^2 \theta - \cot^2 \theta$.		1
(d)	For the parabola $y = 5 + 2x - x^2$,		
	i. Find the equation of the axis of symmetry.		1
	ii. Find the coordinates of the vertex.		1
	iii. Find the focal length.		1
	iv. Sketch the parabola.		

Question 6	(12 Marks)	Commence a NEW page.	Marks
(a)	What is the equation of the line through $(-1, 3)$ and perpendicular to the line $5x - 2y - 3 = 0$?		2
(b)	For $y = 2x^3 - 3x^2 - 12x + 2$		
	i. Find all stationary points and determine their nature.		2
	ii. Find the point(s) of inflexion.		2
	iii. Sketch the curve.		2
(c)	Find the volume generated when the area between the curve $y = x^3$ and the line $y = x$ where $x \geq 0$ is rotated about the x axis.		4

- | | | |
|------------------------------|----------------------|--------------|
| Question 7 (12 Marks) | Commence a NEW page. | Marks |
|------------------------------|----------------------|--------------|
- (a) Inflation is still increasing, but with various policies to reduce inflation over the last few years had an effect. Draw a graph to represent this information with I (inflation) against t (time). **2**
- (b) Show that $\frac{\cos \alpha}{1 + \sin \alpha} + \frac{1 + \sin \alpha}{\cos \alpha} = \frac{2}{\cos \alpha}$. **2**
- (c) Draw the graph of $y = e^{-x}$. By drawing another graph on the same set of axes, show that $f(x) = e^{-x} - x + 1$ has exactly one root. **2**
- (d) If A is $(1, 3)$, B is $(5, 6)$ and P is (x, y)
- i. Find the lengths PA and PB in terms of x and y . **2**
 - ii. Find the locus of the points whose distance from A is double its distance from B . **2**
- (e) The p -th term of a certain geometric progression is c times the first term. **2**
- Show that the common ratio r is given by

$$\log r = \frac{\log c}{p - 1}$$

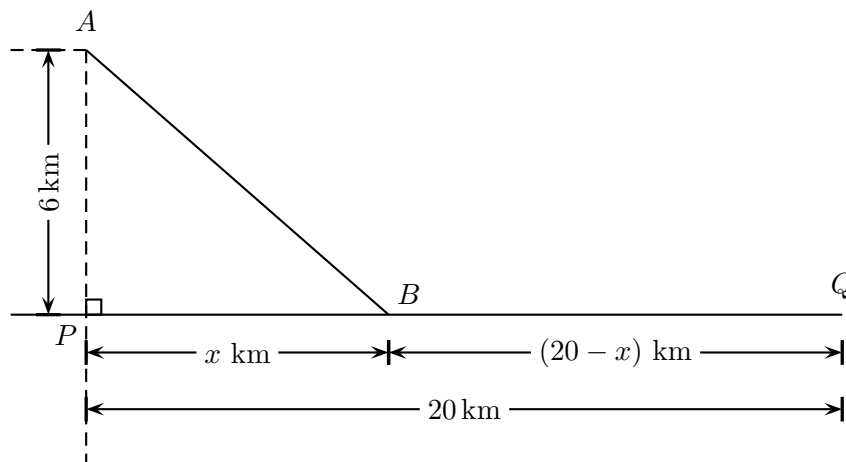
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|------------------------------|----------------------|--------------|
| Question 8 (12 Marks) | Commence a NEW page. | Marks |
|------------------------------|----------------------|--------------|
- (a) The series $7 + 13 + 31 + \cdots$ has $3^n + 4$ as its n -th term. Write down
- i. the seventh term. **1**
 - ii. the sum of the first n terms. **2**
- (b) Prove that the line $4x - y - 8 = 0$ is a tangent to the curve $x^2 = 2y$ and find the point of contact. **3**
- (c) Solve the following where $0 \leq x \leq 2\pi$:
- i. $(2 \cos x + 1)(\cos x - 1)$. **2**
 - ii. $3 \tan^2 x = 1$. **2**
 - iii. $2 \sin \left(x - \frac{\pi}{3}\right) = -1$. **2**

Question 9 (12 Marks)

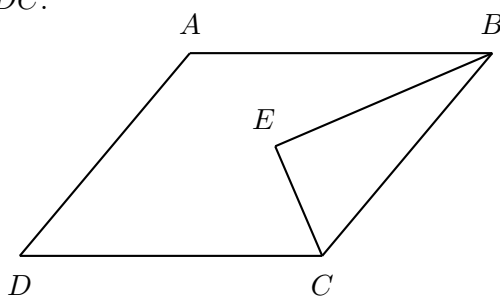
Commence a NEW page.

Marks

- (a) A courier on a bike is in open country at the point A and is 6 km from the nearest point P on a straight sealed road. He wishes to get as quickly as possible to a point Q on the sealed road 20 km from P . If the maximum speed across the open country is 40 km/h and along the sealed road is 50 km/h,



- i. Write an expression for the times taken to traverse AB and BQ , if B is a point on the sealed road between P and Q . 2
 - ii. Write an expression for the total time T for the journey. 2
 - iii. Find the distance from P that he should find the road. 2
- (b) Find the volume of revolution if the curve $y = \sec x$ from $x = 0$ to $x = \frac{\pi}{4}$ is rotated about the x axis. 3
- (c) In the figure, $ABCD$ is a parallelogram (not to scale). EB bisects $\angle ABC$, $\angle ADC = 64^\circ$ and $EC \perp DC$.



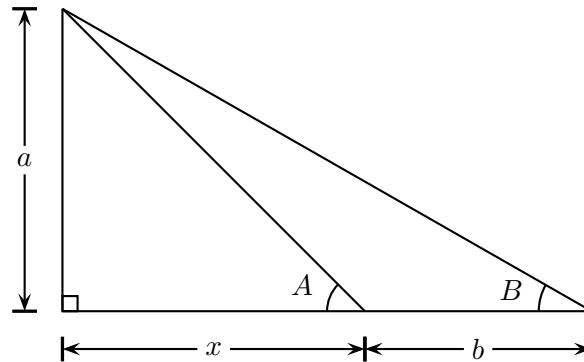
- i. Draw a diagram showing all the information given. 1
- ii. Calculate the size of $\angle BEC$, giving reasons. 2

Question 10 (12 Marks)

Commence a NEW page.

Marks

- (a) Evaluate $\tan \frac{\pi}{3} + \operatorname{cosec} \frac{\pi}{4}$. **1**
- (b) i. Show that $\frac{1}{2x-3} - \frac{1}{2x+3} = \frac{6}{4x^2-9}$. **2**
- ii. Hence find $\int \frac{dx}{4x^2-9}$. **2**
- (c) From the diagram, $\tan A - \tan B = 1$.



- i. Prove that $x^2 + bx - ab = 0$. **2**
- ii. If $a = 3$ and $b = 6$, find x in simplest surd form. **2**
- iii. Prove that $\tan A = \frac{\sqrt{3}+1}{2}$ and find $\tan B$ in similar form. **2**
- iv. Find A and B to the nearest degree. **2**

End of paper.

Q1

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$$(i) \quad 2\sqrt{11} \times 5\sqrt{11} = 10 \times 11 \\ = 110 \quad \checkmark\checkmark$$

$$(ii) \quad x^2 - 3x - 7 = 0 \quad \therefore x = \frac{3 \pm \sqrt{9 - 4(1)(-7)}}{2} \\ = \frac{3 \pm \sqrt{37}}{2} \quad \checkmark \\ \doteq 4.54138, -1.54138 \\ = 4.54, -1.54 \quad (2.dp) \quad \checkmark\checkmark$$

$$(iii) \quad f(-1) + f(0) + f(2) \\ = (-1-2) + 0^2 + 2^2 \quad \checkmark \\ = -3 + 4 \\ = 1 \quad \checkmark$$

$$(iv) (a) \quad \sin \frac{7\pi}{6} = -\sin \frac{\pi}{6} \\ = -\frac{1}{2} \quad \checkmark$$

$$(b) \quad \log_2 1 = 0 \quad \checkmark$$

$$(v) (a) \quad A = \frac{1}{\sqrt{2}+1} \cdot \frac{1}{\sqrt{2}-1} \\ = \frac{1}{2-1} \\ = 1 \text{ cm}^2 \quad \checkmark$$

$$(b) \quad P = 2 \cdot \frac{1}{\sqrt{2}+1} + 2 \cdot \frac{1}{\sqrt{2}-1} \\ = \frac{2}{\sqrt{2}+1} \cdot \frac{\sqrt{2}-1}{\sqrt{2}-1} + \frac{2}{\sqrt{2}-1} \cdot \frac{\sqrt{2}+1}{\sqrt{2}+1} \\ = 2\sqrt{2} - 2 + 2\sqrt{2} + 2 = 4\sqrt{2} \quad \checkmark\checkmark$$

$$(2) \text{ i) } \frac{d}{dx} (4x^{-5} + 2x) \\ = 4 - 2x^{-2} \\ = 4 - \frac{2}{x^2} \quad \frac{2}{2}$$

$$\text{b) } \frac{d}{dx} (e^{3x^2-5}) = 6x e^{3x^2-5} \quad \frac{2}{2}$$

$$\text{c) } \frac{d}{dx} (x^3 e^{-x}) = 3x^2 e^{-x} - e^{-x} x^3 \\ = x^2 e^{-x} (3-x) \quad \frac{2}{2}$$

$$\text{d) } \frac{1}{\sin x} \times \cos x = \cot x \quad \frac{2}{2}$$

[cot x is worth 1 mark]

$$\text{ii) } y = (3x-2)^4 \\ y' = 12(3x-2)^3 \\ y'' = 108(3x-2)^2 \quad 1$$

As y'' is a square, it will never change sign through the derivative $\therefore$ no inflexion $\frac{2}{2}$

$$\text{iii) } -5 \leq 3-2x \leq 5$$

$$-8 \leq -2x \leq 2$$

$$4 \geq x \geq -1 \quad \frac{2}{2}$$

$$\text{iv) } -1 \leq x \leq 4$$

Accepted: $x \geq -1$ and $x \leq 4$.

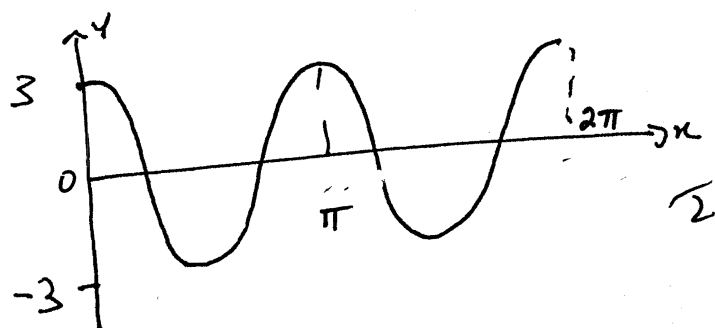
2(ii) The second mark is for a valid explanation. i.e.

$y'' \geq 0$ for all real $x \therefore$ no change in concavity.

OR even testing one point before + one point after.

iii) $p = \frac{2\pi}{2} = \pi$

$a = 3$



iv) $y = k(x+1)(x-2)$

vi) a) $\ln(x^2+1) + c$

b) $2\ln x + 5e^x + c$

c) $\frac{1}{2} \ln(e^{2x} + 4) + c$

4) $V = 2000 - 40t + 0.2t^2$

a) $t = 0$ $V = 2000 \text{ cm}^3$

b) either solve quadratic to get $t = 100$
or $\frac{dV}{dt} = 0$ to get $t = 100$

c) $\frac{dV}{dt} = .4t - 40$

when $t = 30$

$\frac{dV}{dt} = -28$

Decreasing at rate of $28 \text{ cm}^3/\text{s}$

v) $S_{\infty} = \frac{a}{1-r}$
 $a = 8$
 $r = \frac{1}{2}$
 $S_{\infty} = \frac{8}{1-\frac{1}{2}} = \frac{16}{1}$

vi) x | -1 | -1/2 | 0 | 1/2 | 1 |
f(x) | 1 | $\sqrt{3/2}$ | $\sqrt{2}$ | $\sqrt{5/2}$ | $\sqrt{3}$ |

$A \doteq \frac{1}{6} [\sqrt{2} + 1 + 4\sqrt{3/2}] + \frac{1}{6} [\sqrt{2} + \sqrt{3} + 4\sqrt{5/2}]$
 $\doteq 2.80$ (20p)
(only 2.80 for last mark).

vii) $\frac{\frac{1}{4}}{\frac{3}{4}} = \frac{1}{3}$

viii) $y' = 4x + 4$

|| when $4x + 4 = 8$ $\xrightarrow{\text{worth 1}}$
 $4x = 4$
 $x = 1$

14 at $(1, -5)$

(5) i) $y' = x \cos x + \sin x$

when $x = \frac{\pi}{2}$ $y' = 1$ $y = \frac{\pi}{2}$

∴ Normal is

$$y - \frac{\pi}{2} = -1 \left(x - \frac{\pi}{2} \right)$$

$$y - \frac{\pi}{2} = -x + \frac{\pi}{2}$$

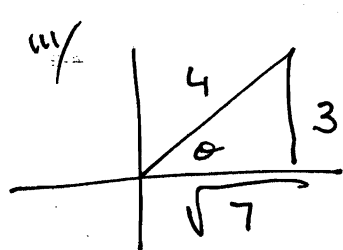
$$x + y = \pi$$

ii) a) $2 + k = 5$

$$k = 3$$

b) $3x = 12$

$$k = 4$$



a) $\tan \theta = \frac{3}{\sqrt{7}}$

b) $\sec \theta = \frac{4}{\sqrt{7}}$

c) $\cos^2 \theta - \sin^2 \theta = 1$

iii) $y = -x^2 + 2x + 5$

a) $x = -\frac{b}{2a} = \frac{-2}{-2}$

$$x = 1$$

b) vertex is $(1, 6)$

c) $y - 5 = -x^2 + 2x$

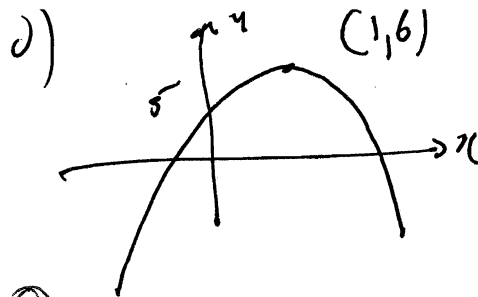
$$-y + 5 + 1 = x^2 - 2x + 1$$

$$-y + 6 = (x - 1)^2$$

$$(x - 1)^2 = -1(y - 6)$$

$$4a = 1$$

$$a = \frac{1}{4}$$



(6) i) $2x + 5y + k = 0$ (a.m. = $-\frac{2}{5}$)

through $(-1, 3)$

$$-2 + 15 + k = 0$$

$$k = -13$$

$$2x + 5y - 13 = 0$$

ii) $y = 2x^3 - 3x^2 - 12x + 2$

$$y' = 6x^2 - 6x - 12$$

$$y'' = 12x - 6$$

a) start when $y' = 0$

$$6x^2 - 6x - 12 = 0$$

$$(x - 2)(x + 1) = 0$$

$$x = -1 \text{ or } x = 2$$

$$y'' < 0$$

max at

$$(-1, 9)$$

$$y'' > 0$$

min at

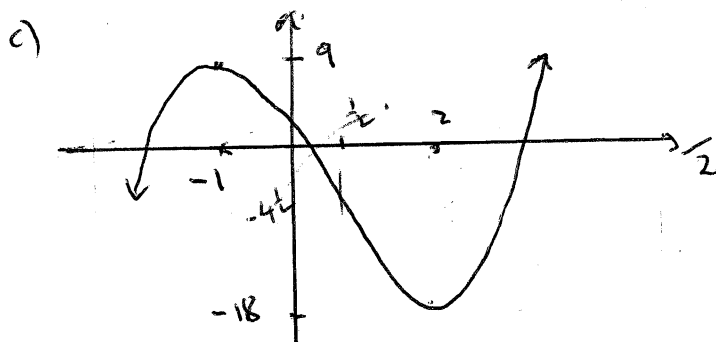
$$(2, -18)$$

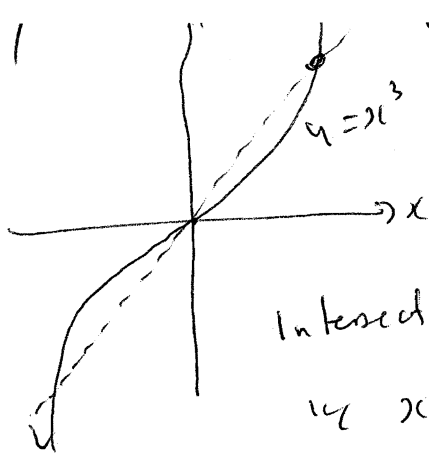
b) inflection when $y'' = 0$

sign changes

$$x = \frac{1}{2}$$

test $f''(\frac{1}{2} - \epsilon) -ve$ } sign change
 $f''(\frac{1}{2} + \epsilon) +ve$ } ∴ inflection at $(\frac{1}{2}, -4\frac{1}{2})$





Intersect where $x^3 = x$ ✓

∴ $x = 0, x = 1$

$$V = \pi \int_0^1 y^2 dx$$

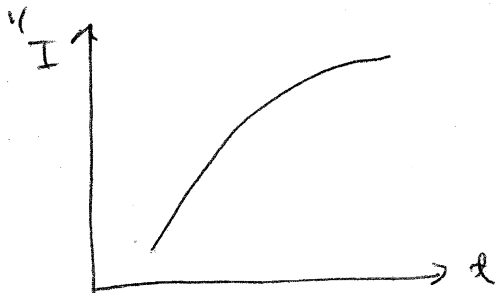
$$= \pi \int_0^1 x^2 dx - \pi \int_0^1 x^3 dx$$

$$= \pi \left[\frac{x^3}{3} \right]_0^1 - \pi \left[\frac{x^4}{4} \right]_0^1$$

$$= \frac{\pi}{3} - \frac{\pi}{4}$$

$$= \frac{\pi}{12}$$

⑦



1 - concave
1 - concavity

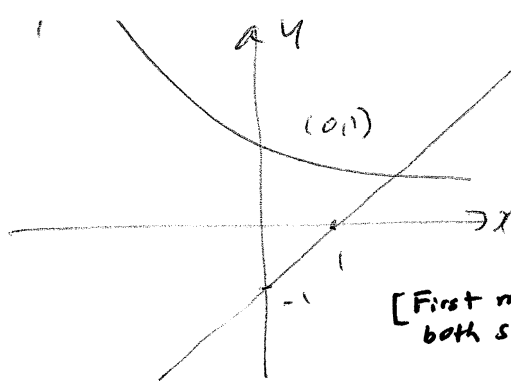
$$11) LHS = \frac{\cos^2 x + 1 + 2 \sin x + \sin^2 x}{\cos x (1 + \sin x)}$$

$$= \frac{2 + 2 \sin x}{\cos x (1 + \sin x)}$$

$$= \frac{2 (1 + \sin x)}{\cos x (1 + \sin x)}$$

$$= \frac{2}{\cos x}$$

$$= \sec x$$



[First mark for both shapes] ✓

$e^{-x} - x + 1 = 0$. Because 1 point of intersection, 1 root. ✓
other graph $y = x - 1$

$$14) a) PA = \sqrt{(x-1)^2 + (y-3)^2}$$

$$PB = \sqrt{(x-5)^2 + (y-6)^2}$$

$$b) PA = 2 PB$$

$$PA^2 = 4 PB^2$$

$$(x-1)^2 + (y-3)^2 = 4[(x-5)^2 + (y-6)^2]$$

$$x^2 - 2x + 1 + y^2 - 6y + 9 = 4(x^2 - 10x + 25 + y^2 - 12y + 36)$$

$$3x^2 - 38x + 3y^2 - 42y + 23 = 0$$

$$15) T_p = ar^{p-1}$$

$$\therefore Ca = ar^{p-1}$$

$$c = ar^{p-1}$$

$$\log c = \log ar^{p-1}$$

$$\log r (p-1) = \log c$$

$$\log r = \frac{\log c}{p-1}$$

18) $y = 3^n + 4$

a) $T_n = 3^n + 4$

b) $S_n = \sum_{k=1}^n (3^k + 4)$
 $= \frac{3(3^n - 1)}{3 - 1} + 4n$
 $= \frac{3(3^n - 1)}{2} + 4n$

ii) $y = 4x - 8 \wedge x^2 = 2y$

where $x^2 = 8x - 16$

$x^2 - 8x + 16 = 0$

$(x - 4)^2 = 0$

$x = 4$

$y = 8$

$(4, 8)$ is 1 point of intersection

$\therefore$ line is a tangent.

iii) a) $\cos x = -\frac{1}{2}$ or $\cos x = 1$

$\begin{array}{c|c} s & A \\ \hline + & C \end{array}$ Base $x = \frac{\pi}{3}$

$x = 0, 2\pi$

$\begin{array}{c|c} s & A \\ \hline + & C \end{array}$ $x = \frac{2\pi}{3}, \frac{4\pi}{3}$

b) $\tan^2 x = \frac{1}{3}$

$\tan x = \pm \frac{1}{\sqrt{3}}$

$\begin{array}{c|c} s & A \\ \hline + & C \end{array}$

$x = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$

c) $\sum (x - \frac{\pi}{3}) = -\frac{1}{2}$

$\begin{array}{c|c} s & A \\ \hline + & C \end{array}$ Base $x = \frac{\pi}{6}$
 $x - \frac{\pi}{3} = \frac{7\pi}{6} \sim \frac{11\pi}{6}$



$x = \frac{7\pi}{6} + \frac{\pi}{3} \sim \frac{11\pi}{6} + \frac{\pi}{3}$

$= \frac{9\pi}{6}$

$\frac{13\pi}{6}$

$= \frac{3\pi}{2}$

$\frac{17\pi}{6}$

for F. + D. min.

$$q(a)(i) \quad T = \frac{D}{S}$$

$$T_{AB} = \frac{\sqrt{20^2 + 36}}{40} \quad \checkmark$$

Accepted in
minutes.

$$T_{BQ} = \frac{20-x}{50} \checkmark$$

(b) $T = \frac{\sqrt{x^2 + 36}}{40} + \frac{20 - x}{50}$ ✓ (1)

$$(c) \quad \frac{dT}{dx} = \frac{\frac{1}{2}(x^2+36)^{-\frac{1}{2}} 2x}{40} - \frac{1}{50}$$

$$= \frac{x}{40\sqrt{x^2+36}} - \frac{1}{50} \checkmark$$

$$\text{let } \frac{x}{40\sqrt{x^2+36}} - \frac{1}{50} = 0$$

$$\frac{x}{40\sqrt{x^2+36}} = \frac{1}{50}$$

$$5x = 4\sqrt{x^2 + 36}$$

$$25x^2 = 16(x^2 + 36)$$

$$9x^2 = 16.36$$

$$x^2 = 16.4$$

$$x = 8 \quad (x > 0) \checkmark$$

$$f'(8 - \varepsilon) \neq 0$$

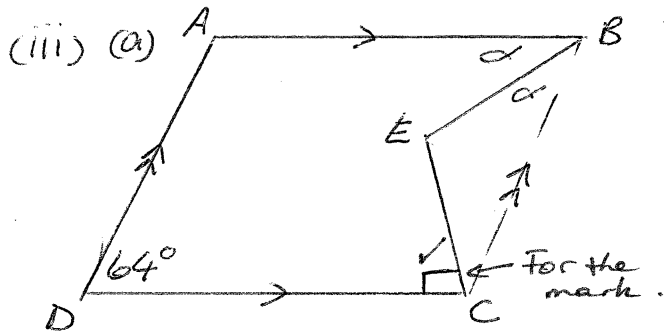
$$f'(z) = 0$$

$$f'(z+\varepsilon) \not\geq 0$$

$\therefore$ Max at $x=8$ ✓

Min

(ii) (d) $V = \pi \int_0^{\frac{\pi}{4}} \sec^2 x \, dx$ ✓
 $= \pi [\tan x]_0^{\frac{\pi}{4}}$ ✓ (3)
 $= \pi u^3$ ✓



$$\begin{aligned} \text{(b) } \angle ECB &= 180 - (90 + 64) \\ &\text{(oint. } \angle\text{'s, parallel lines)} \\ &= 26^\circ \end{aligned}$$

$$\angle ABC = 64^\circ \text{ (opp. } \angle \text{'s of parm)}$$

$$\angle EBC = 32^\circ \text{ (data)}$$

$$\therefore \angle BEC = 180 - (32 + 26)$$
$$(\angle \text{sum of } \triangle)$$
$$= 122^\circ \quad \checkmark \quad \checkmark$$

If right angle not between EC and DC, question much easier (3)

$\frac{2}{3}$ km Junior

$$\sqrt{36+x^2}$$

Q10

$$(i) \tan \frac{\pi}{3} + \operatorname{cosec} \frac{\pi}{4} = \sqrt{3} + \sqrt{2}$$

✓

$$(ii) \quad \frac{1}{2x-3} - \frac{1}{2x+3} = \frac{2x+3 - 2x-3}{(2x-3)(2x+3)}$$

(a)

$$= \frac{6}{4x^2-9}$$

✓

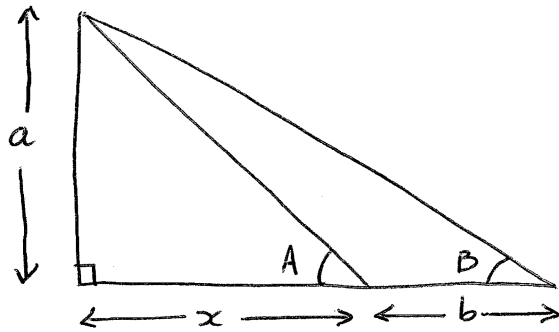
$$(b) \int \frac{dx}{4x^2-9} = \frac{1}{6} \int \frac{6}{4x^2-9} dx$$
$$= \frac{1}{6} \int \left(\frac{1}{2x-3} - \frac{1}{2x+3} \right) dx$$
$$= \frac{1}{12} \ln(2x-3) - \frac{1}{12} \ln(2x+3) + C$$
$$= \frac{1}{12} \ln \left(\frac{2x-3}{2x+3} \right) + C$$

✓

✓

Q10 (cont'd)

(iii)



Given:

$$\tan A - \tan B = 1$$

(a) $\tan A - \tan B = 1$

$$\therefore \frac{a}{x} - \frac{a}{b+x} = 1$$

$$\therefore \frac{ab+ax-ax}{x(x+b)} = 1$$

$$\therefore ab = x(x+b)$$

$$\therefore x^2 + bx - ab = 0$$

(b)

$$x^2 + 6x - 18 = 0$$

$$x = \frac{-6 \pm \sqrt{36 - 4(1)(-18)}}{2}$$

$$= \frac{-6 \pm \sqrt{108}}{2} = \frac{-6 \pm 6\sqrt{3}}{2}$$

$$= -3 \pm 3\sqrt{3}$$

[But $x > 0 \therefore x = -3 + 3\sqrt{3}$]

(c) $\tan A = \frac{a}{x} = \frac{3}{-3+3\sqrt{3}} = \frac{1}{\sqrt{3}-1}$
 $= \frac{\sqrt{3}+1}{2}$

$$\tan B = \frac{3}{6+(-3+3\sqrt{3})} = \frac{3}{3+3\sqrt{3}} = \frac{1}{1+\sqrt{3}}$$

$$= \frac{\sqrt{3}-1}{2}$$

(d) $A \doteq 53.79^\circ, B \doteq 20.104^\circ$

ie $A = 54^\circ$ (nearest degree)
 $B = 20^\circ$